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Differences among lychee cultivars in responses to *Peronophythora litchii* implicate cell wall and carbohydrate remodeling

Abstract: Litchi downy blight, caused by the oomycete *Peronophythora litchii* C.C. Chen ex W.H. Ko, H.S. Chang, H.J. Su, C.C. Chen & L.S. Leu, damages lychee (*Litchi chinensis* Sonn.) leaves, flowers and fruit. We reanalysed five public RNA-sequencing cohorts to identify cultivar-dependent infection responses and test their transfer across tissues and times. Discovery in Guiwei and Yurong1 leaves tested 19 445 expressed genes. Two hundred and six genes passed genome-wide testing, mappability and gene-model quality control; 16 remained stable under alternative quantification, statistical analysis, expression filtering, leave-one-library-out analysis and mapping checks. Twelve internally robust genes formed the highest supported tier and included candidates involved in sugar transport, starch metabolism, cell-wall remodelling, lipid signalling and RNA metabolism. In an independent pericarp time course, LITCHI005518, encoding an alpha-xylosidase, and LITCHI001963 showed same-direction cultivar-dependent effects, whereas five genes reversed direction. A frozen 206-gene signature changed between leaf and fruit and across infection times. Most of 18 earlier candidates responded within at least one cultivar, but none passed the genome-wide cultivar-interaction criterion. These results distinguish general infection responses from cultivar-dependent effects and prioritise testable targets for experimental validation.

Keywords: oomycete disease; host response; differential expression; transcript usage; reproducible bioinformatics; crop improvement

INTRODUCTION

Lychee (*Litchi chinensis* Sonn.) is an important subtropical fruit crop. Litchi downy blight, caused by *Peronophythora litchii*, affects young leaves, inflorescences and fruit, and cultivars differ in disease response. Published work has described pathogen virulence machinery, host ethylene disruption and cultivar-specific early transcriptional programmes (Sun et al. 2017; Sun et al. 2019; Li et al. 2023). However, most RNA-sequencing studies analyse a single cohort, so prominent candidates can reflect tissue, time or sampling context rather than stable cultivar differences.

A cultivar-by-infection interaction asks whether the infection response differs between cultivars; it is not equivalent to testing infection within each cultivar. We first explored Guiwei

and Yurong1 leaves and nominated 18 defence-related genes. We then registered a separate genome-wide analysis before computing confirmatory outcomes. Dataset roles, models, thresholds, internal robustness gates, external evaluations and evidence tiers were fixed in advance.

Here we integrate five public cohorts to identify quality-controlled cultivar-dependent responses, evaluate their robustness and transport across biological contexts, and distinguish supported, null and opposite-direction outcomes. We also assess transcript usage, pathway and promoter evidence, and provide a reproducible candidate resource for targeted validation.

MATERIAL AND METHODS

Study design and data. The exploratory stage preceded registration. The confirmatory protocol was frozen on 18 August 2026 and amendments were time-stamped. Five public cohorts had non-interchangeable roles (Table 1; Figure 1A): PRJNA830488/GSE201243 was the discovery cohort; PRJNA450886 was the primary cross-context evaluation; PRJNA922966/GSE222651 tested generic infection transfer; PRJNA922965/GSE222650 provided small-RNA data; and PRJNA1090613/GSE262200 remained exploratory because time and resistance metadata were unresolved. All inferences are conditional on independence of deposited libraries, which could not be verified from source-tree, pooling or extraction metadata.

Preprocessing and discovery. The host SCAU_Lch_v2.0 assembly (Hu et al. 2022) and pathogen reference were combined. Reads were trimmed with fastp (Chen et al. 2018), aligned with STAR (Dobin et al. 2013), counted with featureCounts (Liao et al. 2014) and independently quantified with Salmon (Patro et al. 2017). Libraries required at least 10 million surviving read pairs and 40% uniquely aligned reads. DESeq2 fitted gene counts as cultivar + treatment + cultivar:treatment (Love et al. 2014). Genes with counts of at least 10 in three libraries were tested; candidates required genome-wide Benjamini-Hochberg $q < 0.05$ and absolute interaction $\log_2$ fold change at least $\log_2(1.5)$ (Benjamini & Hochberg 1995). Exon mappability was assessed with GenMap (Pockrandt et al. 2020).

Robustness and external evaluation. Frozen genes were retested using Salmon gene-level counts, edgeR quasi-likelihood inference (Robinson et al. 2010), a counts-per-million filter and every leave-one-library-out refit, followed by an observed-mapping check. The primary external model estimated the Heiye-minus-Guiwei difference in infection response at 24 h in pericarp. Passing results required family-adjusted $q < 0.05$, absolute $\log_2$ fold change at least

log₂(1.5), a confidence interval excluding zero and discovery-direction agreement; significant opposite-direction effects were recorded separately.

Additional analyses. A frozen signed score summarised the 206 genes across external contrasts. Pathways were mapped through one-to-one *Oryza* orthologues to Plant Reactome (Naithani et al. 2024). Transcript usage was tested with DRIMSeq and stageR (Nowicka & Robinson 2016; Van den Berge et al. 2017). Annotation used Swiss-Prot DIAMOND matches, InterPro and *Oryza* orthology (Buchfink et al. 2015; Jones et al. 2014). Promoter analysis tested JASPAR plant motifs against expression- and GC-matched backgrounds (Rauluseviciute et al. 2024); FIMO provided sensitivity testing (Grant et al. 2011). The PmiREN archive was audited before small-RNA integration (Guo et al. 2020).

Evidence integration and reproducibility. Deterministic rules combined discovery, mapping, internal robustness, primary external evidence and orthogonal evidence without additive scoring. Tier A required all classes; Tier B required discovery and complete internal robustness with partial or non-testable external evidence; Tier C was reserved for a robust, externally supported pathway. Snakemake workflows enforced staged analysis and SHA-256 manifests (Mölder et al. 2021). Complete outputs, source data and exact software versions are supplied in the supplementary archive.

RESULTS

Discovery quality and genome-wide interaction. All 12 discovery libraries passed technical gates: read survival was 99.56-99.66%, unique alignment 84.44-86.22% and Salmon mapping 80.77-82.56% (Figure 1B). Principal components separated cultivar and infection status (Figure 1C). Among 19 445 expressed genes, 262 met the registered interaction threshold; uniform mappability and gene-model control retained 206 candidates (Figure 2A,C). Recomputing adjustment only among 13 602 QC-eligible genes recovered all 206, showing that post-test quality control produced a conservative subset.

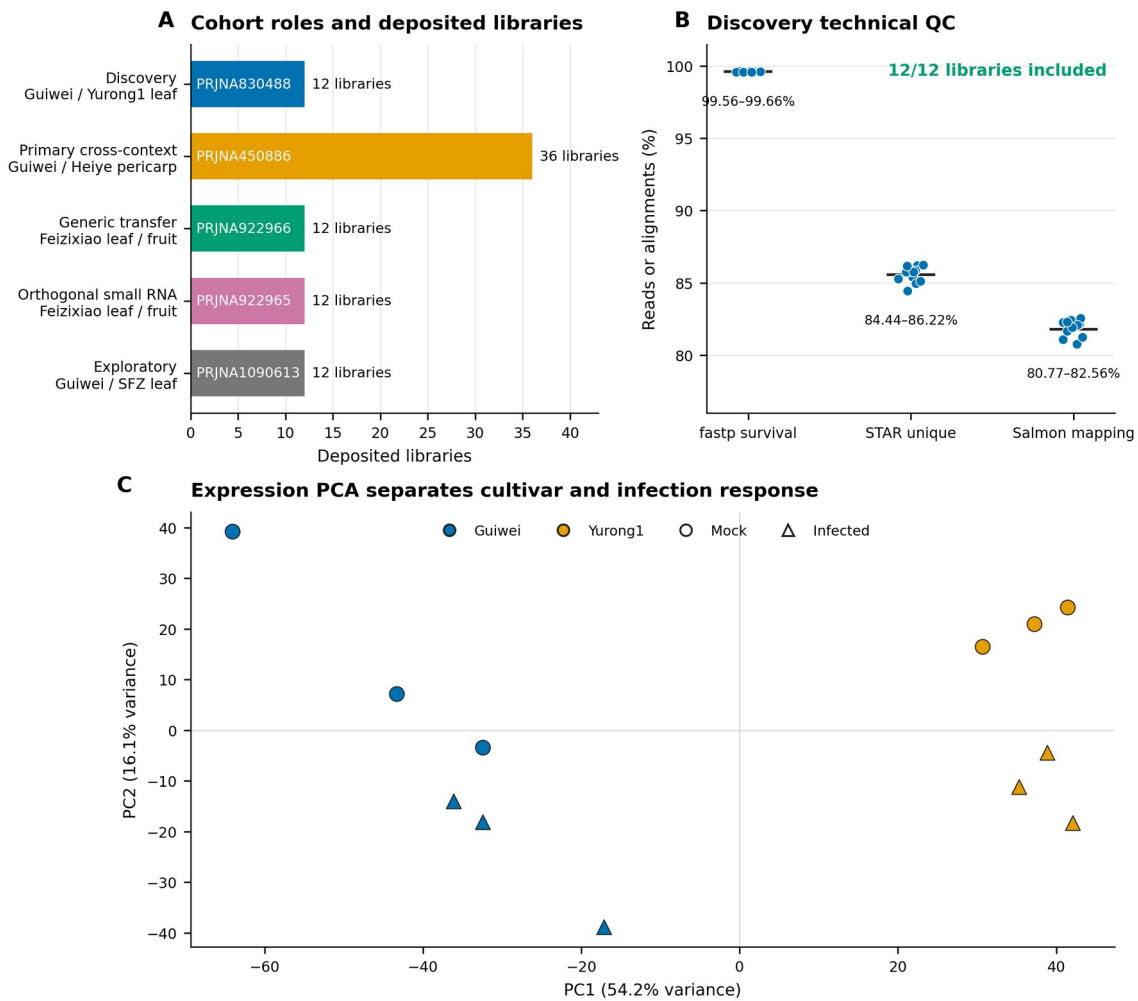


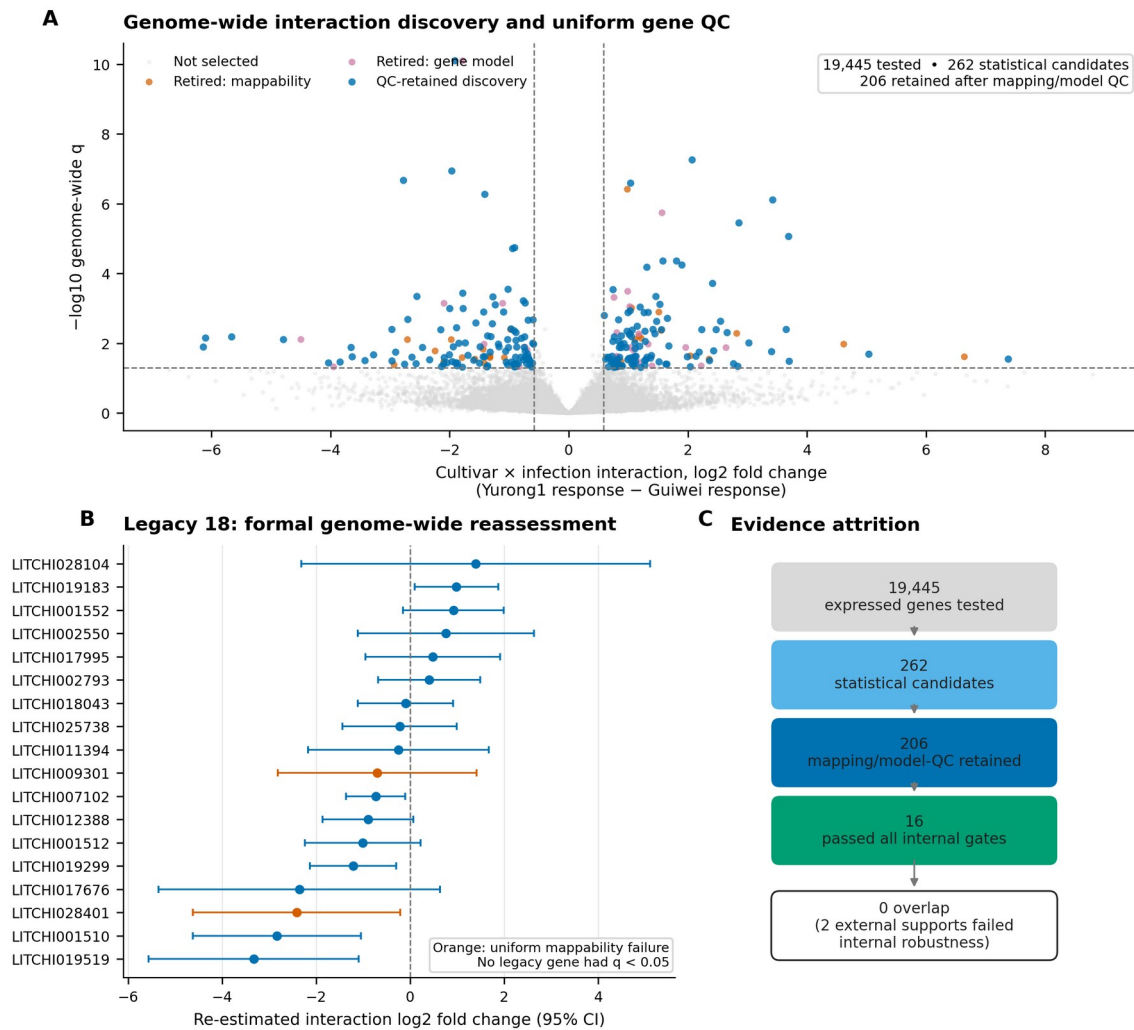
Figure 1. Study cohorts and discovery quality. (A) Five public cohorts with fixed roles. (B) Per-library read survival, STAR unique alignment and Salmon mapping; all 12 discovery libraries passed. (C) Principal components separate cultivar and infection status.

Table 1. Fixed dataset roles and eligibility.

Accession	Design	Fixed role
PRJNA830488 / GSE201243	Guiwei and Yurong1 leaf; 24 h	Discovery
PRJNA450886	Guiwei and Heiye pericarp; 6, 24, 48 h	Primary external
PRJNA922966 / GSE222651	Feizixiao leaf and fruit; 24 h	Generic transfer
PRJNA922965 / GSE222650	Feizixiao small RNA; 24 h	Orthogonal modality
PRJNA1090613 / GSE262200	Guiwei and SFZ leaf	Exploratory only

Exploratory candidates and internal robustness. Thirteen of 18 earlier candidates were infection responsive in Guiwei and 16 in Yurong1 at genome-wide $q < 0.05$, but none met the separate cultivar-interaction criterion (Figure 2B). Two failed uniform mappability. Of the 206 frozen genes, 196 passed independent quantification, 19 passed the edgeR gate, 205 passed expression filtering and 125 passed all leave-one-out refits. Eighteen passed these four gates jointly and 16

98 remained after observed-mapping checks (Figure 3A). DESeq2 and edgeR interaction estimates
99 correlated at $r = 0.995$ genome-wide.



100
101 Figure 2. Genome-wide discovery and reassessment of exploratory candidates. (A) Interaction effects across 19
102 445 genes; 262 met the statistical threshold and 206 remained after quality control. (B) Interaction estimates for
103 18 exploratory candidates. (C) Attrition to the internally robust set.

104 Cross-context outcomes. In the primary pericarp evaluation, 184 candidates were measurable.
105 LITCHI005518 and LITCHI001963 met every criterion in the discovery direction, while five
106 genes met magnitude and significance criteria in the opposite direction (Table 2; Figure 3C).
107 Discovery and external effects were uncorrelated (Pearson $r = -0.05$), and 95 of 184 shared
108 direction. The 16 internally robust genes and two cross-context-supported genes did not
109 overlap; for sets of these sizes, the probability of zero overlap under independence is 0.85
110 (Figure 3B).

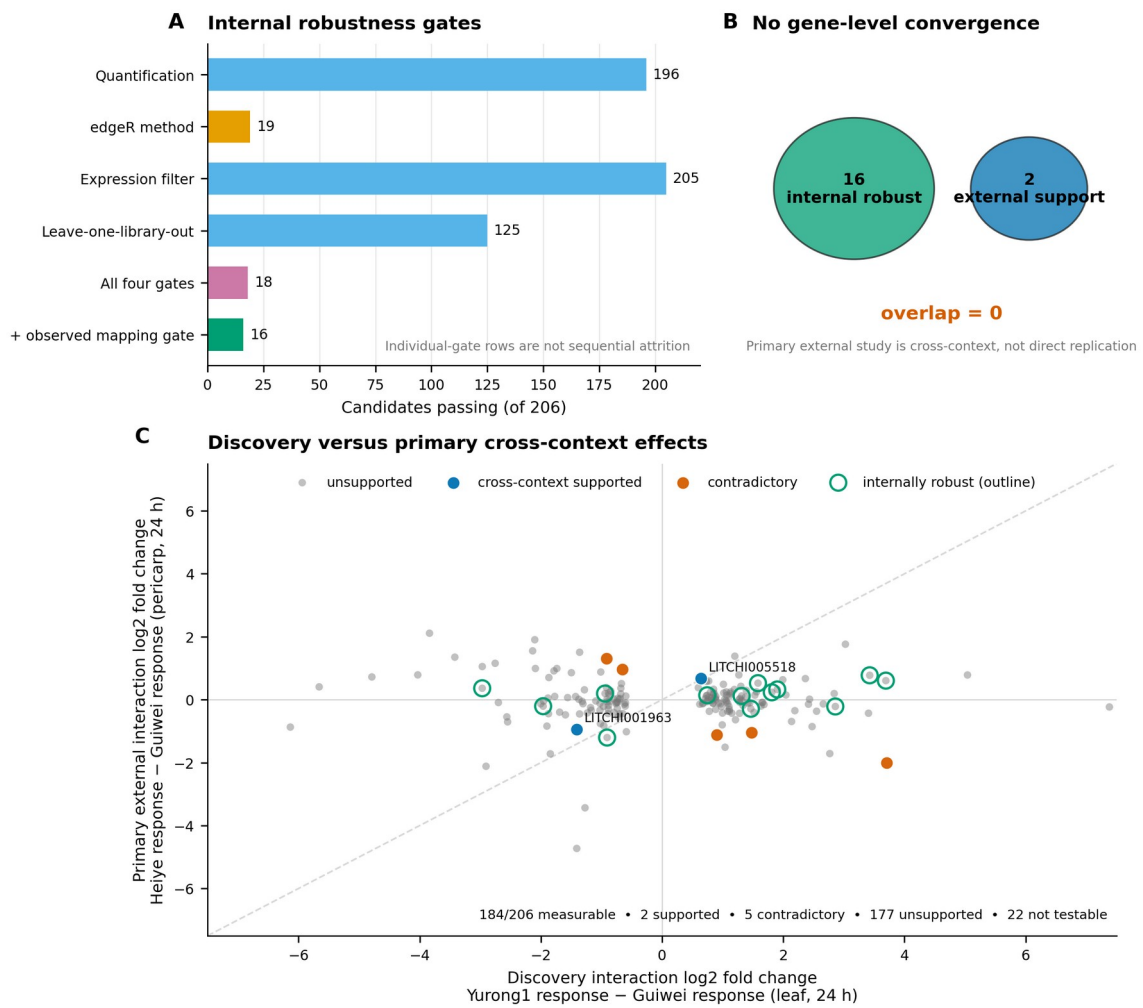


Figure 3. Internal robustness and primary external evaluation. (A) Retention by each robustness gate. (B) The 16 internally robust and two cross-context-supported genes do not overlap. (C) Discovery effects against 24-h pericarp effects; the diagonal is a visual reference.

Table 2. Supported and opposite-direction genes in the primary external evaluation. Effects are cultivar-by-infection interaction log2 fold changes; external intervals are 95% confidence intervals.

Gene	Annotation	Discovery log2FC	Discovery q	External log2FC (95% CI)	External q	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterised; 1:1 Oryza orthologue	-1.41	0.024	-0.94 (-1.43 to -0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	-0.92	0.040	+1.31 (+0.78 to +1.85)	0.0002	Reversed
LITCHI029534	Thiamine thiazole synthase 2	+1.48	0.002	-1.04 (-1.61 to -0.48)	0.010	Reversed
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	-0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Reversed
LITCHI008313	Beta-glucosidase 44	+0.90	0.045	-1.12 (-1.81 to -0.44)	0.034	Reversed
LITCHI001700	Fe(II)/2-OG-dependent dioxygenase 4	+3.70	0.033	-2.00 (-3.24 to -0.77)	0.035	Reversed

Signatures, pathways and transcript usage. The frozen 206-gene score tracked infection positively in leaf (0.107, $q = 0.0015$) and negatively in fruit (-0.181, $q < 0.001$), with a significant tissue interaction (0.289, $q < 0.001$). In pericarp it changed from 0.109 at 24 h to -0.240 at 48 h ($q = 0.043$), demonstrating tissue and temporal dependence (Figure 4B). Six pathways were discovered, but none passed primary external criteria; circadian rhythm alone passed all internal gates (Figure 4A). Conditional analysis identified 225 transcript-usage events across 152 genes; none passed the primary cross-context threshold (Figure 5).

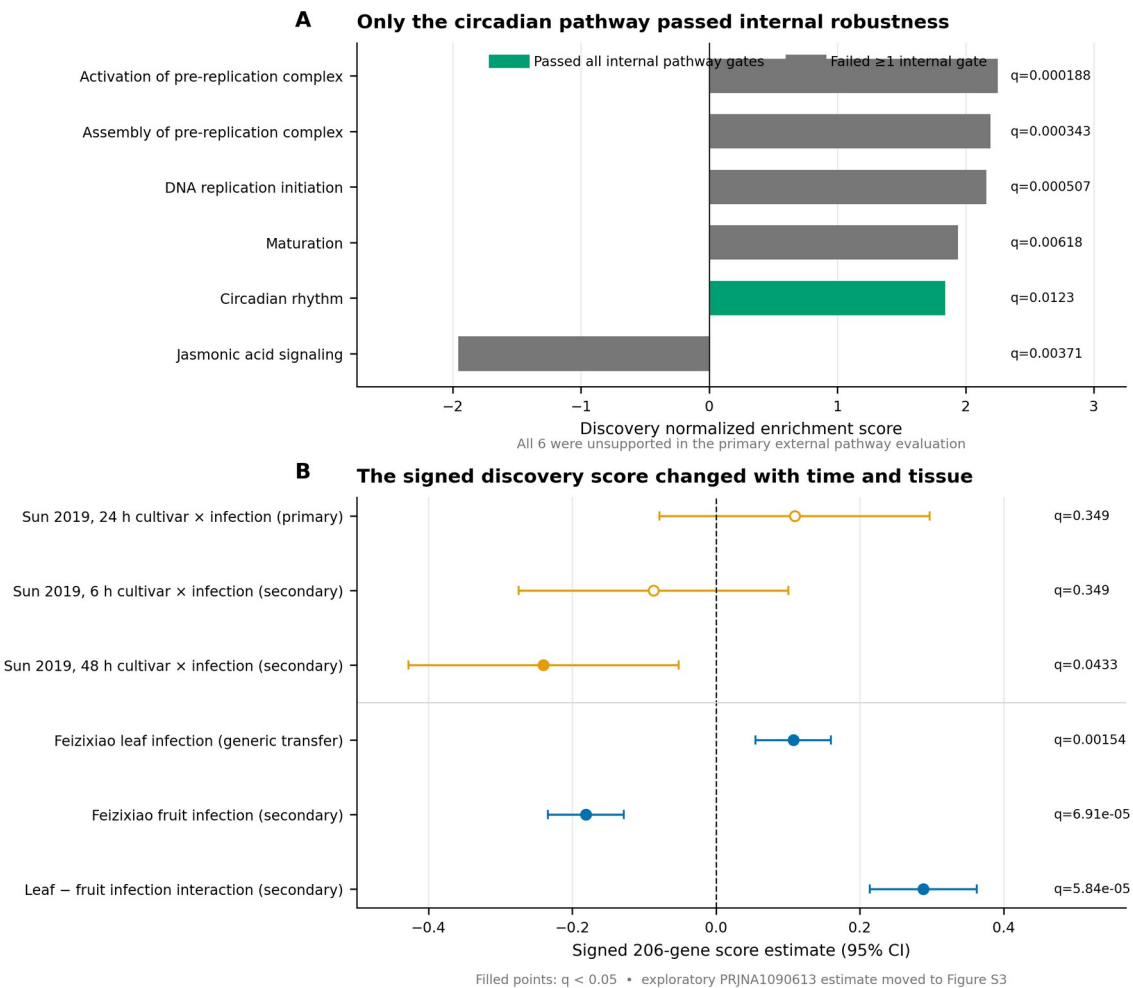


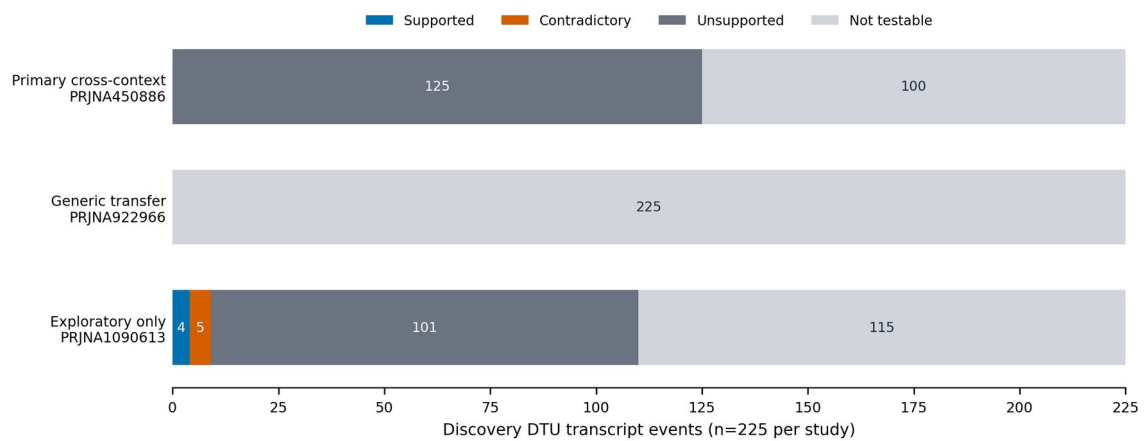
Figure 4. Pathway and signature evaluation. (A) Six discovery pathways; circadian rhythm passed all internal gates, but none passed external criteria. (B) The frozen 206-gene signature across six non-exploratory external contrasts; filled points denote $q < 0.05$.

A Conditional DTU discovery



Stage-wise q values and signed usage-interaction differences

B External DTU follow-up remained separated by study role



No primary event was supported; exploratory outcomes remain non-confirmatory.

Figure 5. Conditional transcript usage. (A) Discovery gate and 225 events across 152 genes. (B) External follow-up separated by prespecified study role.

Orthogonal evidence and tiers. Family-level labels were assigned to 150 candidates. No motif passed the registered 80-of-100 matched-background rule, and small-RNA coherence was not testable because PmiREN had no exact Litchi entry (Figure 6A,B). The frozen signature used apegglm-shrunken weights (Zhu et al. 2019), and pathway robustness included competitive testing with camera (Wu & Smyth 2012). Evidence integration produced no Tier A gene or Tier C pathway, placed 12 internally robust genes in Tier B, retained 191 entities for exploratory follow-up and excluded 65 through predefined quality or direction rules (Figure 6C). Tier B included genes linked to sugar transport, starch and cell-wall metabolism, membrane and lipid transport, chaperone regulation, RNA metabolism and specialised metabolism.

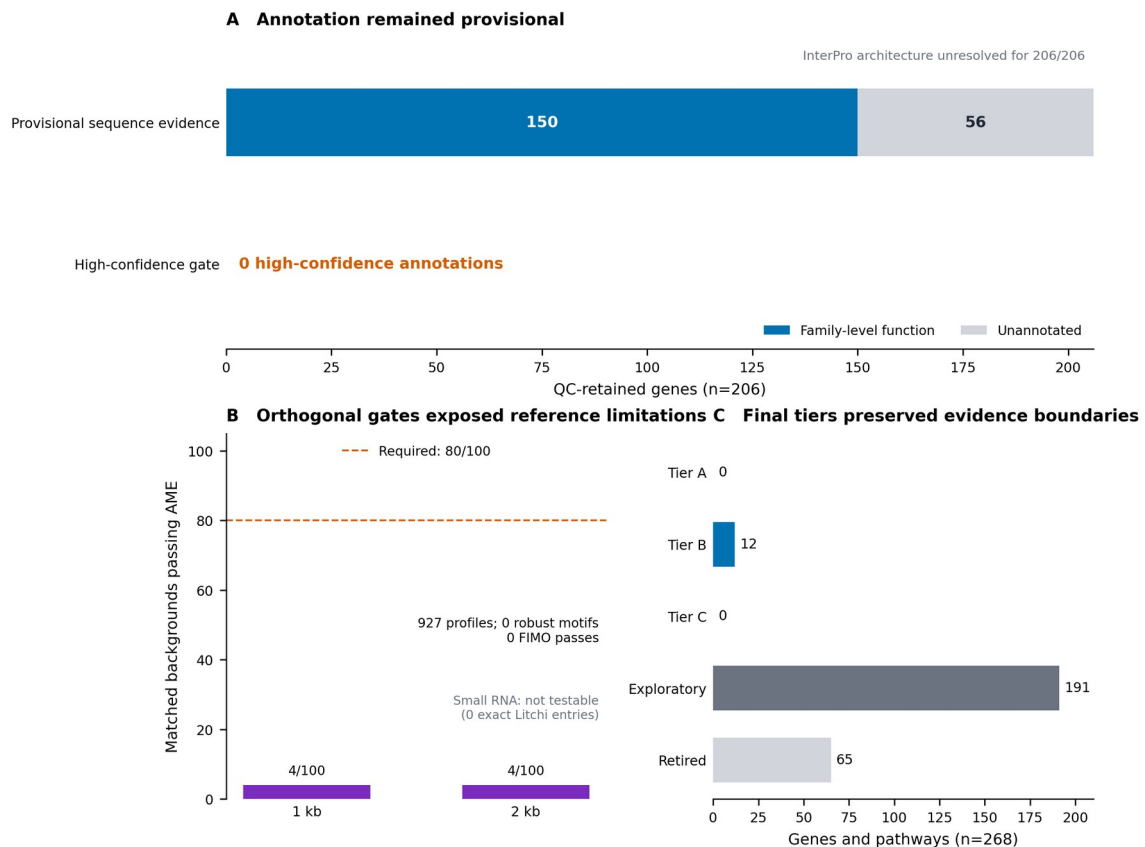


Figure 6. Annotation, orthogonal evidence and prioritisation. (A) Annotation of 206 candidates. (B) Motif robustness and small-RNA reference gate. (C) Deterministic tiers over 268 frozen entities.

DISCUSSION

The analysis separates infection responsiveness from cultivar-dependent response. Most exploratory genes responded within at least one cultivar, yet none passed the genome-wide interaction criterion. This is not a contradiction: a strong response within a cultivar can coexist with an imprecise difference between cultivars, especially with three deposited libraries per cell. Mapping control also identified two signals that could not be assigned unambiguously to one locus.

The prioritised genes point to cell-wall and carbohydrate dynamics. Tier B contains a phosphomannomutase/phosphoglucomutase, a starch-branching enzyme, an ERD6-like sugar transporter and a serine carboxypeptidase; the cross-context set includes alpha-xylosidase LITCHI005518, which acts on xyloglucan. This convergence is consistent with previous lychee work emphasising sugar metabolism and wall reinforcement (Sun et al. 2019). Other Tier B annotations implicate DIR1-like lipid transfer, BAG-family chaperone regulation and RNA metabolism (Berka et al. 2022); these remain hypotheses rather than demonstrated resistance mechanisms.

The external results show that cultivar-associated responses depend strongly on tissue and time. Two genes transported across different cultivar and tissue contrasts, whereas five reversed direction and the broader signature changed between leaf and fruit and between 24 and 48 h. Thus below-threshold or opposite-direction external outcomes should not be described as failed technical replication; they delimit the contexts in which a response may be useful.

Limitations follow from the public designs. Biological independence of deposited libraries is undocumented; inoculum, developmental stage and wounding metadata are incomplete; one reference assembly was used; and no same-tissue, same-time independent cohort was available. None of the candidates has been perturbed experimentally. Independent infections, qPCR and functional analysis of LITCHI005518 and the Tier B genes are therefore required before causal or breeding claims.

CONCLUSION

A registered multi-cohort analysis identified 206 quality-controlled cultivar-interaction candidates, a stable core of 16 genes, 12 internally robust Tier B genes and two genes supported across an independent tissue and cultivar contrast. The results implicate cell-wall and carbohydrate remodelling while showing that broader responses vary with tissue and infection time. The released evidence tables and workflows provide a transparent basis for targeted experimental validation rather than causal claims.

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